DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

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1. TITLE	2. IDENTIFICATION NUMBER
MASTER INSTRUMENT LIST (MIL)	DI-MISC-81338

3. DESCRIPTION / PURPOSE

3.1 Master Instrument List (MIL) Data File is an inventory list of instruments and control system components that are adjustable and permanently fixed or attached to an associated systems or equipment and are used to monitor, measure or control operations, functional performances parameters or physical quantities.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
930701	SH/05J4		

7. APPLICATION / INTERRELATIONSHIP

7.1 This Data Item Description contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 NAVSEA Publication S9040-AA01DX-010/SWBS; Ship Work Breakdown Structure may be obtained from: Commander, Naval Sea Systems Command, Attn: SEA 04TD4L5, 2531 Jefferson Davis Highway, Arlington, VA 22242.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		N6928

10. PREPARATION INSTRUCTIONS

10.1 Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.
10.2 Format. The Master Instrument List (MIL) shall be presented in the standard American Standard Code for Information Interchange (ASCII) format on 5 1/2" high density computer diskettes. Hard copy shall be in a format similar to that shown in Table 1.
10.3 Computer Program. The computer software used shall be capable of conversion to ASCII.
10.4. Content. The diskettes shall contain all the instructions required to format data for review or printout. The instructions shall be menu driven such that the user may select the data items to be presented.
10.4.1 Title Page. The title shall be prominently displayed on the title page. The preparing orgainzation shall be identified. Letter size of the title and preparing orgainzation shall be consistent with the requirements of 10.3 of this DID.

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11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 10. Preparation Instructions (Continued)

10.4.2 Definitions. The next page after the title page shall contain a list of acronyms and their definitions. Page format for acronyms and their definitions shall be similar to that as presented in TABLE II. Additional pages shall be used as required to list all acronyms used throughout the Master Instrument List. The title of this list shall be displayed on all pages used. The list shall be in alphabetical order.

10.4.3 Page Format. Data page format shall be as follows:

10.4.3.1 Retrieval Date. The diskette menu shall include a retrieval date field. This date shall appear in the most right hand field of the first line of each data page.

10.4.3.2 Revision. The latest revision level and date of the Master Instrument List shall appear at the most left hand fields of the first and second lines of each data page. The revision level shall be; Original, Revision A, Revision B, etc..

10.4.3.3 Data Items. The data items to be specified are as follows:

(a) Reference Number
(b) Part Of
(c) Sub System
(d) Location
(e) Sub Location
(f) Model Number
(g) Range
(h) Nomenclature
(i) Function Expanded
(j) Instrument Number
(k) ESWBS Number
(l) ESWBS TITLE
(m) APL Number
(n) NSN
(o) Drawing Number
(p) Technical Manual Number
(q) Set Point
(r) Increase or Decrease
(s) Set Point Action
(t) Reset Point
(u) Reset Action

10.4.3.4 Page Header. Two lines shall be centered at the top of each page: (First Line) HM&E MASTER INSTRUMENT LIST - (Vessel Class). The (Second Line) SORTED BY _____. Up to the first three columns used to sort the data base shall be listed on the SORTED BY line .

10.4.3.5 Header Divider. A dashed line shall appear beneath the data item heading fields across the entire active area of the page.

10.4.3.6 Field Spacing. No less than two character spaces shall be provided between each field allocated for each data item. The field allocation for each data item is defined herein.

10.4.3.7 Page Numbering. Each data page shall be numbered at the bottom right-hand of the page beginning with page 1

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10.4.4 Data Fields The data fields shall be as follows:

10.4.4.1 Reference number (REF #) The (REF #) for data entries shall range between 0001 and 9999. A sequential reference number shall be assigned to each instrument and control system component placed in the list. After assignment, the REF # identifies a unique instrument or control system component and is used only once. Grouping by systems or location is preferential. The REF # field shall not exceed six characters.

10.4.4.2 Part of. The PART OF field defines the system to which the instrument or control system component is attached. System acronyms shall be as shown in TABLE II. The acronyms shall be consistent with those developed by 10.4.6. The PART OF data field shall not exceed 10 characters.

10.4.4.3 Sub system. The SUB SYSTEM field shall be used, when applicable, to define the secondary system which interfaces with the primary system. The SUB SYSTEM field shall not exceed 10 characters.

10.4.4.4 Location. This field defines the location of the instrument or control system component by compartment area. Compartment designations shall be standardized and abbreviated to minimized field usage. The LOCATION field shall not exceed 10 characters. Examples of typical abbreviations are as follows:

a. MMRLL - Main Machinery Room Lower Level
b. EOS - Enclosed Operating Station
c. RDREQRM - Radar Equipment Room
e. FASSTA1 - Fueling at Sea Station 1

10.4.4.5 Location other, When instruments or control system components are not located in specific spaces, the location shall be identified by deck, frame and zone, Examples are as follows:

a. 1-223-0 b. 1-151-1

10.4.4.6 Sub location (SUB LOC). The SUB LOC field describes the approximate mounting location of an instrument or control system component within a compartment. The SUB LOC field shall not exceed 10 characters.

10.4.4.7 Model number (MOD NO). This field shall contain the model number of the instrument or control system component and shall not exceed 15 characters.

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Block 10. Preparation Instructions (Continued)

10.4.4.8 Range. This field shall define the full range capability of the instrument or control system component with the appropriate engineering units. For signal conditioners, the out put voltage of the unit or engineering units shall be defined. For instrument or control system components which measure negative and positive values, the first value given shall be the negative range of the unit followed by a dash-zero-dash (-0-) and then by the positive range in inch units and the pressure range in pounds per square inch, the vacuum range of inches shall be interpreted to be inches of mercury or as specified in the contract. The RANGE field shall not exceed 15 characters.

10.4.4.9 Nomenclature. This field item shall define the instrument or control system component. See Table III for a sample listing. The NOMENCLATURE field shall not exceed 20 characters.

10.4.4.10 Function Expanded (FUNCTION EXPD). This field shall define the parameter being measured by the instrument or control system component identified in the Nomenclature field. The FUNCTION EXPANDED field shall not exceed 45 characters.

10.4.4.11 Instrument Number (INST NO). This field identifies the instrument or control system component number from data on the label plate. The INST NO field shall not exceed 20 characters. Examples are as follows:

a. LPA-GA-1
b. ECW-PS-2

10.4.4.12 Expanded Ship Work Breakdown Structure Number (ESWBS #). This field shall provide the ESWBS # for each instrument or control system component. The ESWBS # field shall not exceed 14 characters.

10.4.4.13. ESWBS Title. This field shall list the ESWBS title in accordance with NAVSEA manual S9040-AA-IDX-010/SWBS. The ESWBS Title field shall not exceed 60 characters.

10.4.4.14. Allowance Parts List Number (APL #). This field shall list the APL#, when available. The APL # field shall not exceed 11 characters.

10.4.4.15 National Stock Number (NSN). This field shall list the NSN, when known, for each instrument or control system component. The NSN field shall not exceed 22 characters.

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10.4.4.16 Drawing number (DWG #). This field shall list the NAVSEA drawing(s) which locate and identify the instrument or control system component. There may be up to five fields and each field shall not exceed 15 characters.

10.4.4.17 Technical manual number (TM #). The TM # fields list the manuals which identify and describe the instrument or control system component. These may be equipment or system manuals, ie, Propulsion Operating Guides (POGs), or Ship Information Books (SIBs). There may be up to five fields and each field shall not exceed 16 characters.

10.4.4.18 Set point. This field identifies the value at which the switch opens or closes. The SET POINT field shall not exceed 15 characters.

10.4.4.19 Increasing or decreasing (INC or DEC). This field identifies INC (I) or DEC (D) switch activation and shall not exceed one character.

10.4.4.20 Set point action (SET PT ACTION). This field identifies whether the switch opens (O) or closes (C) when activated and shall not exceed one character.

10.4.4.21 Reset point. This field identifies the value at which the switch resets and shall not exceed 15 characters.

10.4.4.22 Reset action. This field identifies whether the switch resets in the open (O) or closed (C) position and shall not exceed one character.

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TABLE I. Data format example

HM&E MASTER INSTRUMENT LIST - DD963 CLASS
SORTED BY: PART OF, SUB SYSTEM, LOCATION

REF #	PART OF INST NO DWG #:	SUB SYSTEM	LOCATION ESWBS # TM #:	SUB LOC	MODEL NO ESWBS TITLE SET POINT	RANGE INC OR DEC	NOMENCLATURE SET PT ACTION	RESET POINT	FUNCTION EXPANDED APL # RESET ACTION	NSN
10011	AC1 AC1-PS-1	CW	AUX1UL 5140	4-230-1	222-1118 AIR CONDITIONING SYSTEM 235 DEGF	I	TEMPERATURE SWITCH O	225 DEGF	HIGH DISCHARGE SUMMARY ALARM C	
10041	AC2 AC2-PS-1	CW	AUX1UL 5140	4-240-1	222-1118 AIR CONDITIONING SYSTEM 235 DEGF	I	TEMPERATURE SWITCH O	225 DEGF	HIGH DISCHARGE SUMMARY ALARM C	
10071	AC3 AC3-PS-1	CW	PMPRM2 5140	3-410-1	222-1118 AIR CONDITIONING SYSTEM 235 DEGF		TEMPERATURE SWITCH		HIGH DISCHARGE SUMMARY ALARM C	
10168	BA PA-FM-1	PA	MER1ML 55162		PRAIRIE AIR SYSTEM	0-650SCFM	FLOWMETER		PRAIRIE AIR 610670016	
10303	CRP1 CRP1-GA-10A 203-4539616 893-5181498 893-5181499		MER2LL 2451 0905-LP-552-6010		0-300PSID PROPULLERS AND PROPULSORS 70 PSID	I	IND PRESSURE SWITCH C		ATTACHED PUMP FILTER	
10222	CRP2 CRP2-GA-10A 203-4539616 893-5181498 893-5181499		MER2LL 2451 0905-LP-552-6010		0-300PSID PROPULLERS AND PROPULSORS 70 PSID	I	IND PRESSURE SWITCH C		ATTACHED PUMP FILTER	
10460	DP1		AUX1LL 5311		30-0-30PSI DISTILLING PLANT, FLASH TYPE		COMPOUND GAGE		2 STAGE SHELL	6685-974-6532
10515	EC 509-4539722 532-5386645	SWRDMEQ	5362		0-5UMHO COOLING WATER, ELECTRONICS, DW/SW		CONDUCTIVITY METER		DEMINERALIZER	

FIGURE I SAMPLE FORMAT

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TABLE II. Examples of acronyms versus definitions

AFFF	Aqueous Film Forming Foam	HPAC	High Pressure Air Compressor
CW	Chilled Water	LOS	Lube Oil System
ECW	Electronic Cooling Water	MRG	Main Reduction Gear
EPCC	Enclosed Propulsion Control Center	MSW	Main Seawater System
PW	Potable Water	FSEE	Free Standing Electronic Enclosure
SSDG	Ship Service Diesel Generator		

TABLE III. Instruments and control systems components

The following are examples of equipment which should be listed:

Alarms Threshold Circuitry
Buffers (Analog and Digital)
Converters:
  Frequency to Direct Current (F/DC)
  Direct Current to Frequency (DC/F)
  Analog to Digital (A/D)
  Digital to Analog (D/A)
Feedback Potentiometers
Flow Measuring Devices (liquid & gas)
Multiplexer Channels
Permissive Function Circuitry
Power Supplies for Control and
  Measurement Systems
Display Devices:
  digital indicators
  analog meters
  indicator lamps
Signal Conditioners
Switches:
  pressure/vacuum
  temperature
  flow
  level
Resistance Temperature Detectors (RTD)
Tachometers
Thermocouples
Thermometers
Throttle/Pitch Controls
Torsionmeters
Transducers:
  pressure/vacuum level
  vibration
  accelerometers
  flow
Gauges:
  pressure/vacuum
  temperature
  voltmeters
  wattmeters
Ammeters
Frequency Meters
Conductivity

The above list shall not be considered to be all inclusive.

FIGURE II SAMPLE FORMAT

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